

Project Proposal: Network Protocol Analyzer

Muhammad Junaid Akhtar

Sap ID: 63877

Course: Operating Systems

1.Objective:

To develop a command-line tool that captures network packets in real-time and provides analysis of the protocols, packet counts, and network traffic statistics. The tool will help users monitor network activity, analyze protocol usage, and gather insights from network traffic.

2.Background:

Analyzing network protocols is crucial for network administration, troubleshooting, and security monitoring. Tools like Wireshark and tcpdump provide packet-level visibility, but creating a simplified, real-time analyzer helps users understand protocol traffic patterns without complex interfaces. This project demonstrates packet capture and protocol analysis using Python.

3. Scope:

The Network Protocol Analyzer will:

- Capture live packets from a selected network interface
- Detect protocols (TCP, UDP, ICMP, HTTP, DNS, ARP)
- Display packet summaries (time, IPs, ports, protocol, length)
- Apply filters (protocol, port, IP)
- Maintain traffic statistics (total packets, protocol distribution, top ports)
- Save packets to PCAP files

4. Tools and Libraries:

- **Python 3** – Programming language
- **Scapy** – Packet capture and protocol identification
- **Colorama** – Terminal color output
- **argparse** – Command-line argument parsing
- **datetime, collections** – Packet statistics and tracking

5. Expected Outcomes

- A working CLI-based packet analyzer
- Real-time protocol summaries and statistics
- Exported PCAP files for offline analysis

6. Benefits

- Enables real-time network monitoring and protocol analysis for better traffic visibility.
- Helps identify unusual or suspicious traffic patterns for basic security analysis.
- Provides a practical learning platform for understanding network protocols and packet structures.

7. Limitations

- Command-line based system without a graphical user interface (GUI).
- Supports only basic filtering and does not include advanced alert mechanisms.
- Limited analysis of encrypted traffic and not suitable for large-scale enterprise networks.

8. Conclusion

The proposed project aims to develop a real-time Network Protocol Analyzer capable of capturing network packets, identifying multiple communication protocols, and generating meaningful traffic summaries. The tool will provide users with the ability to monitor network activity, apply protocol-based filters, and analyze packet-level communication efficiently through a command-line interface.